

WSQ Business Transformation with OpenClaw

Lesson Plan (LP)

Course Code: TGS-2022015374

Institution: Tertiary Infotech Pte. Ltd.

Instructor: Dr. Alfred Ang

Duration: 1 Day | Total lab time: 280 minutes

1-Day Schedule

09:00 – 09:20 Welcome, Digital Attendance & Course Overview

09:20 – 09:50 Topic 1 (part): Overview of OpenClaw – Concepts [30 min]

09:50 – 10:20 Lab 1: OpenClaw Hosting (VPS or Docker Desktop) [30 min]

10:20 – 10:35 Break

10:35 – 11:05 Lab 2: OpenClaw Model (MiniMax / OpenAI / Ollama) [30 min]

11:05 – 11:35 Lab 3: OpenClaw Channel (Telegram + WhatsApp) [30 min]

11:35 – 12:05 Topic 2: OpenClaw Applications (demos) [30 min]

12:05 – 12:25 Lab 4: OpenClaw Tools (AgentMail / Browser / Firecrawl) [20 min]

12:25 – 13:25 Lunch

13:25 – 13:45 Lab 5: OpenClaw Skills (ClawHub) [20 min]

13:45 – 14:15 Lab 6: Cron Jobs and Heartbeat [30 min]

14:15 – 14:45 Lab 7: OpenClaw Security (10-step guide) [30 min]

14:45 – 15:05 Lab 8: OpenClaw Dashboard (SSH tunnel for VPS) [20 min]

15:05 – 15:25 Lab 9: OpenClaw Cost Saving [20 min]

15:25 – 16:05 Lab 10: Google Integration (Gmail / Calendar / Chat) [40 min]

16:05 – 16:30 TRAQOM Survey, Certificate, Assessment & Closing

Learning Outcomes

SFw: Emerging Technology Synthesis ICT-SNA-3011-1.1

- LO1: Deploy OpenClaw on a Hostinger VPS or Docker Desktop local machine
- LO2: Connect LLM providers (MiniMax free, OpenAI OAuth, Ollama, Default config)
- LO3: Set up Telegram (BotFather) and WhatsApp (QR pairing) channels
- LO4: Integrate AgentMail, Agent Browser, and Firecrawl tools
- LO5: Install and invoke skills from the ClawHub marketplace
- LO6: Schedule tasks with cron jobs and monitor uptime with heartbeat
- LO7: Apply 10-step security hardening
- LO8: Access the OpenClaw web dashboard (local or SSH tunnel)
- LO9: Optimise costs via model selection and context compaction
- LO10: Integrate Google Workspace via service account

Prerequisites

- Laptop: Windows 10/11 (WSL2 recommended), macOS 12+, or Ubuntu 22.04+
- Docker Desktop installed (for local lab environment)
- Admin / sudo rights on the machine
- Telegram account + phone with WhatsApp installed
- Free MiniMax account at <https://www.minimaxi.com> (no credit card required)
- Free accounts at <https://agentmail.to> and <https://www.firecrawl.dev>
- Google account (personal or Workspace) for Lab 10

Lab Environments – Two Options

Students choose ONE environment for all labs:

- Option A: Hostinger VPS – Ubuntu 22.04 LTS, SSH, 24/7 cloud deployment
- Referral link (discount): <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>
- Option B: Docker Desktop – local machine, no cloud account needed
- Pull image: `docker pull openclaw/openclaw:latest`

Assessment

Written Assessment (Short Answer) + Oral Questioning – Open Book

Practical demonstration – students must show:

- `openclaw gateway status` – Gateway: running
- Agent replies in Telegram or WhatsApp chat
- Firecrawl tool returns page summary
- Cron job fires and sends result to channel
- Dashboard accessible at `http://localhost:18789`
- Google: agent reads Gmail or Calendar event

TRAQOM Survey and Certificate Delivery: mandatory QR code from LMS at end of class.

LMS: <https://ai-lms-tms.tertiaryinfo.tech/>